PHD115 Fundamentals of Chemistry · Chemical Reactions I

Before class: reactions and amounts

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. A periodic table and the answer key are at the back.

1 · Read first (5 min)

What is a chemical equation?

In a **chemical reaction**, substances change into new substances. We write it as a **chemical equation**: the **reactants** go on the left of the arrow and the **products** on the right, for example $H_2 + O_2 \rightarrow H_2O$. Labels show the state: (s) solid, (ℓ) liquid, (g) gas, (aq) dissolved in water.

Atoms are never lost

Matter is neither created nor destroyed, so every atom on the left must appear on the right. We balance an equation by putting numbers called **coefficients** in front of formulas: $2H_2 + O_2 \rightarrow 2H_2O$. We never change the small subscripts, because that would change the substance itself.

Reactions have patterns

Many reactions follow a few patterns. Some **join** substances, some **break** them, and some **burn** in oxygen. In a **displacement** reaction, atoms or ions **swap partners**, like dancers holding hands at a party. You will meet all five types in class.

Counting by weighing

Atoms are far too small to count one at a time. Chemists use a counting unit, the **mole**, just as a baker uses a dozen. You already know how to add up atomic masses (formula mass in IC1, molecular mass in IC2). In this lecture those numbers become grams per mole, so you can weigh out exact amounts, which is the everyday work of a pharmacist.

2 · Readiness check

Eleven quick questions on what you need for today.

Q1 CR1-LO1

In $H_2 + O_2 \rightarrow H_2O$, the reactants are:

A. H_2O only

B. H_2 and O_2

C. H_2 and H_2O

D. O_2 only

Q2 CR1-LO1

In a chemical reaction, atoms can be created or destroyed.

True / False

Q3 CR1-LO1

How many oxygen atoms are shown in $2H_2O$?

Q4 CR1-LO1

How many hydrogen atoms are shown in $3NH_3$?

A. 3

B. 4

C. 6

D. 9

Q5 CR1-LO1

Match each state label to its meaning.

Options: solid · liquid · gas · dissolved in water

(s) _____

(ℓ) _____

(g) _____

(aq) _____

Q6 CR1-LO1

Why can't we balance $H_2 + O_2 \rightarrow H_2O$ by changing it to H_2O_2?

A. H_2O_2 is a different substance (hydrogen peroxide)

B. It would have too many electrons

C. Subscripts can only be even numbers

D. It would not be a gas

Q7 CR1-LO3

Think of a compound as two dancers holding hands. In A + BC → AC + B, what happens?

A. A takes C's hand, and B is left alone

B. A and B join to make one big group

C. BC breaks apart and nothing else happens

D. Nothing changes

Q8 CR1-LO5

Recap: the molecular mass of water, H_2O (H = 1.01, O = 16.00).

_____

Q9 CR1-LO5

Recap: the formula mass of NaCl (Na = 22.99, Cl = 35.45).

_____

Q10 CR1-LO4

Recap from IC1: when a sodium atom forms Na^+, it:

A. gains one electron

B. loses one electron

C. gains one proton

D. loses one proton

Q11 CR1-LO5

A dozen always means 12 items. A mole likewise always means the same fixed number of particles.

True / False

3 · Think ahead

No wrong answers here. Make a guess, then check it after the lecture (Parts 1 and 7).

Q12 CR1-LO6

An antacid tablet reacts with stomach acid. Why would a pharmaceutical scientist need a **balanced** equation to work out how much acid one tablet can neutralize?

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Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 5.0–5.6 and 6.0–6.5. Single and double displacement reactions adapted from *Beginning Chemistry* (Ball), LibreTexts, CC BY-NC-SA 3.0, section 4.3. Shared under the same licence for non-commercial teaching.

Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 **H** 1.01 2.1																	2 **He** 4.00 —
3 **Li** 6.94 1.0	4 **Be** 9.01 1.5											5 **B** 10.81 2.0	6 **C** 12.01 2.5	7 **N** 14.01 3.0	8 **O** 16.00 3.5	9 **F** 19.00 4.0	10 **Ne** 20.18 —
11 **Na** 22.99 0.9	12 **Mg** 24.31 1.2											13 **Al** 26.98 1.5	14 **Si** 28.09 1.8	15 **P** 30.97 2.1	16 **S** 32.07 2.5	17 **Cl** 35.45 3.0	18 **Ar** 39.95 —
19 **K** 39.10 0.8	20 **Ca** 40.08 1.0	21 **Sc** 44.96 1.3	22 **Ti** 47.87 1.5	23 **V** 50.94 1.6	24 **Cr** 52.00 1.6	25 **Mn** 54.94 1.5	26 **Fe** 55.84 1.8	27 **Co** 58.93 1.9	28 **Ni** 58.69 1.9	29 **Cu** 63.55 1.9	30 **Zn** 65.38 1.6	31 **Ga** 69.72 1.6	32 **Ge** 72.63 1.8	33 **As** 74.92 2.0	34 **Se** 78.97 2.4	35 **Br** 79.90 2.8	36 **Kr** 83.80 —
37 **Rb** 85.47 0.8	38 **Sr** 87.62 1.0	39 **Y** 88.91 1.2	40 **Zr** 91.22 1.4	41 **Nb** 92.91 1.6	42 **Mo** 95.95 1.8	43 **Tc** [98] 1.9	44 **Ru** 101.07 2.2	45 **Rh** 102.91 2.2	46 **Pd** 106.42 2.2	47 **Ag** 107.87 1.9	48 **Cd** 112.41 1.7	49 **In** 114.82 1.7	50 **Sn** 118.71 1.8	51 **Sb** 121.76 1.9	52 **Te** 127.60 2.1	53 **I** 126.90 2.5	54 **Xe** 131.29 —
55 **Cs** 132.91 0.7	56 **Ba** 137.33 0.9	57–71	72 **Hf** 178.49 1.3	73 **Ta** 180.95 1.5	74 **W** 183.84 1.7	75 **Re** 186.21 1.9	76 **Os** 190.23 2.2	77 **Ir** 192.22 2.2	78 **Pt** 195.08 2.2	79 **Au** 196.97 2.4	80 **Hg** 200.59 1.9	81 **Tl** 204.38 1.8	82 **Pb** 207.2 1.9	83 **Bi** 208.98 1.9	84 **Po** [209] 2.0	85 **At** [210] 2.2	86 **Rn** [222] —
87 **Fr** [223] 0.7	88 **Ra** [226] 0.9	89–103	104 **Rf** [267] —	105 **Db** [268] —	106 **Sg** [269] —	107 **Bh** [270] —	108 **Hs** [277] —	109 **Mt** [278] —	110 **Ds** [281] —	111 **Rg** [282] —	112 **Cn** [285] —	113 **Nh** [286] —	114 **Fl** [289] —	115 **Mc** [290] —	116 **Lv** [293] —	117 **Ts** [294] —	118 **Og** [294] —

	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Lanthanides	57 **La** 138.91 —	58 **Ce** 140.12 —	59 **Pr** 140.91 —	60 **Nd** 144.24 —	61 **Pm** [145] —	62 **Sm** 150.36 —	63 **Eu** 151.96 —	64 **Gd** 157.25 —	65 **Tb** 158.93 —	66 **Dy** 162.50 —	67 **Ho** 164.93 —	68 **Er** 167.26 —	69 **Tm** 168.93 —	70 **Yb** 173.05 —	71 **Lu** 174.97 —
Actinides	89 **Ac** [227] 1.1	90 **Th** 232.04 1.3	91 **Pa** 231.04 1.4	92 **U** 238.03 1.4	93 **Np** [237] —	94 **Pu** [244] —	95 **Am** [243] —	96 **Cm** [247] —	97 **Bk** [247] —	98 **Cf** [251] —	99 **Es** [252] —	100 **Fm** [257] —	101 **Md** [258] —	102 **No** [259] —	103 **Lr** [262] —

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 CR1-LO1

In $H_2 + O_2 \rightarrow H_2O$, the reactants are:

A. H_2O only

B. H_2 and O_2

C. H_2 and H_2O

D. O_2 only

Reactants are written to the left of the arrow.

Q2 CR1-LO1

In a chemical reaction, atoms can be created or destroyed.

True / False **→ False**

Atoms are only rearranged. This is the law of conservation of matter.

Q3 CR1-LO1

How many oxygen atoms are shown in $2H_2O$?

2

The coefficient 2 multiplies the whole formula: 2 × 1 O = 2.

Q4 CR1-LO1

How many hydrogen atoms are shown in $3NH_3$?

A. 3

B. 4

C. 6

D. 9

3 molecules × 3 H each = 9.

Q5 CR1-LO1

Match each state label to its meaning.

(s) → solid; (ℓ) → liquid; (g) → gas; (aq) → dissolved in water

Q6 CR1-LO1

Why can't we balance $H_2 + O_2 \rightarrow H_2O$ by changing it to H_2O_2?

A. H_2O_2 is a different substance (hydrogen peroxide)

B. It would have too many electrons

C. Subscripts can only be even numbers

D. It would not be a gas

Changing a subscript changes the substance. Only coefficients may change.

Q7 CR1-LO3

Think of a compound as two dancers holding hands. In A + BC → AC + B, what happens?

A. A takes C's hand, and B is left alone

B. A and B join to make one big group

C. BC breaks apart and nothing else happens

D. Nothing changes

A swaps in for B. This is called a **single displacement** reaction.

Q8 CR1-LO5

Recap: the molecular mass of water, H_2O (H = 1.01, O = 16.00).

18.02 u

2 × 1.01 + 16.00 = 18.02 u.

Q9 CR1-LO5

Recap: the formula mass of NaCl (Na = 22.99, Cl = 35.45).

58.44 u

22.99 + 35.45 = 58.44 u.

Q10 CR1-LO4

Recap from IC1: when a sodium atom forms Na^+, it:

A. gains one electron

B. loses one electron

C. gains one proton

D. loses one proton

Losing an electron will soon be called oxidation.

Q11 CR1-LO5

A dozen always means 12 items. A mole likewise always means the same fixed number of particles.

True / False → **True**

One mole is 6.022×10^{23} particles, whatever they are.

3 · Think ahead

Q12 CR1-LO6

An antacid tablet reacts with stomach acid. Why would a pharmaceutical scientist need a **balanced** equation to work out how much acid one tablet can neutralize?

The balanced equation gives the fixed **ratio** in which the antacid and the acid react (for example $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$: one to two). Using the tablet's mass, its molar mass and this mole ratio, you can calculate exactly how much acid it neutralizes. An unbalanced equation would give the wrong ratio and the wrong answer.

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